

IAD-GPT: Advancing Visual Knowledge in Multimodal Large Language Model for Industrial Anomaly Detection

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Abstract—The robust causal capability of multimodal large language models (MLLMs) holds the potential of detecting defective objects in industrial anomaly detection (IAD). However, most traditional IAD methods lack the ability to provide multimodal human-machine dialogs and detailed descriptions, such as the color of objects, the shape of an anomaly, or specific types of anomalies. At the same time, methods based on large pretrained models have not fully stimulated the ability of large models in anomaly detection tasks. In this article, we explore the combination of rich text semantics with both image-level and pixel-level information from images and propose IAD-GPT, a novel paradigm based on MLLMs for IAD. We employ abnormal prompt generator (APG) to generate detailed anomaly prompts for specific objects. These specific prompts from the large language model (LLM) are used to activate the detection and segmentation functions of the pretrained visual-language model (i.e., CLIP). To enhance the visual grounding ability of MLLMs, we propose text-guided enhancer (TGE), wherein image features interact with normal and abnormal text prompts to dynamically select enhancement pathways, which enables language models to focus on the specific aspects of visual data, enhancing their ability to accurately interpret and respond to anomalies within images. Moreover, we design a multimask fusion (MMF) module

to incorporate mask as expert knowledge, which enhances the LLM's perception of pixel-level anomalies. Extensive experiments on MVTec-AD and VisA datasets demonstrate our state-of-the-art performance on self-supervised and few-shot anomaly detection and segmentation tasks, such as MVTec-AD and VisA datasets. The codes are available at <https://github.com/LiZeWen1225/IAD-GPT>

Index Terms—Few-shot anomaly detection, multimodal large language model (MLLM), self-supervised anomaly detection.

I. INTRODUCTION

THE goal of industrial anomaly detection (IAD) tasks is to identify defects in general objects that differ from normal patterns, such as scratches on leather, damaged capsules, and so on. The application of anomaly detection in industry ensures the smooth progress of production processes and plays a crucial role in monitoring, maintaining, and optimizing industrial production processes.

The research on IAD tasks [1], [2], [3], [4], [5], [6], [7] is constantly developing and making good progress. Current mainstream methods [2], [3], [5], [6], [7] for IAD include feature embedding-based methods [2], [3], [5] and reconstruction-based methods [6], [7]. However, traditional IAD methods are currently limited to providing anomaly detection and segmentation results for objects. These approaches all rely on manually setting thresholds and lack the capability to offer detailed insights into the nature and specifics of detected defects. Meanwhile, image-text matching is often used to detect anomalies in large pretrained model-based approaches like WinCLIP [8], which uses a compositional prompt ensemble based on text templates using generic descriptions of normal/abnormal as text. This has been followed by other researchers in subsequent studies [2], [9], [10], but the method does not fully activate the capability of the large pretrained model. Filo [11], [12] proposes an adaptively learned fine-grained description that leverages domain-specific knowledge to introduce detailed anomaly descriptions, replacing generic normal and abnormal descriptions. Research progress on large language models (LLMs) has been rapid recently. Due to their excellent language understanding and reasoning abilities after large-scale data training, LLMs, such as ChatGPT [13] and Llama [14], have proven their ability to perform translation, paraphrasing, and instruction following tasks in zero sample tasks. In the research of multimodal LLMs (MLLMs) [15], [16], [17], it

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is found that other modal information can be mapped to the feature space of LLMs through fine-tuning. LLM can also understand the information contained in other modalities and make explanations for it. AnomalyGPT [2] is the first to introduce LLMs into IAD and proposes the task of anomaly perception in MLLMs (APMLLMs). MLLMs for anomaly detection eliminate the problem of manually setting thresholds in traditional methods and make the results of IAD and localization more interpretable. However, AnomalyGPT simply fine-tunes image features into LLM through a linear layer, feeds predicted masks as expert knowledge into LLM, and finally allows LLM to make judgments on image anomalies.

In this article, we propose IAD-GPT, which is designed to enhance the efficiency and accuracy of anomaly detection in industrial quality inspection. This method not only supports multiturn human-machine dialogs, allowing operators to delve into potential anomalies through interactive question-and-answer (QA) sessions, but also leverages advanced LLMs to directly analyze anomalies within images without relying on preset threshold values for anomaly detection. Traditional anomaly detection methods typically employ fixed threshold standards: if the detected anomaly value exceeds a certain threshold, the image is flagged as containing an anomaly; otherwise, it is considered normal. In contrast, our approach offers greater flexibility and adaptability by making precise judgments based on specific contexts and using LLMs to directly output intuitive results. Consequently, this method holds significant potential for practical application in production environments, providing a novel perspective and solution for industrial quality inspection.

Fig. 1 shows the difference between our IAD-GPT and previous research. To address the issue of insufficient stimulation of large pretrained model segmentation ability in the compositional prompt ensemble method [8], we employ APG to extend and enrich the semantic content of text prompts. These prompts are used to activate the detection and segmentation capabilities of a pretrained visual-language model, i.e., CLIP [18]. Specifically, we leverage GPT's existing knowledge of most objects in the text domain and use a QA format to generate possible anomaly categories for each object class. These generated texts will serve as one of the key factors in identifying anomalies. To enable the LLM to fully perceive image information, we designed two modules at the image level and pixel level, respectively: text-guided enhancer (TGE) and multimask fusion (MMF). TGE enhances the LLM's anomaly perception capability at the image level by interacting image features with normal/abnormal text prompts to achieve dynamic path selection. Meanwhile, the MMF uses the differences in image-text features across multiple levels to further improve the LLM's anomaly perception capability at pixel level.

Our contributions are summarized as follows.

- 1) We introduce a novel framework named IAD-GPT, via leveraging rich visual knowledge for IAD. Compared with previous IAD methods, IAD-GPT enhances the capability to perceive anomalies beyond traditional approaches.

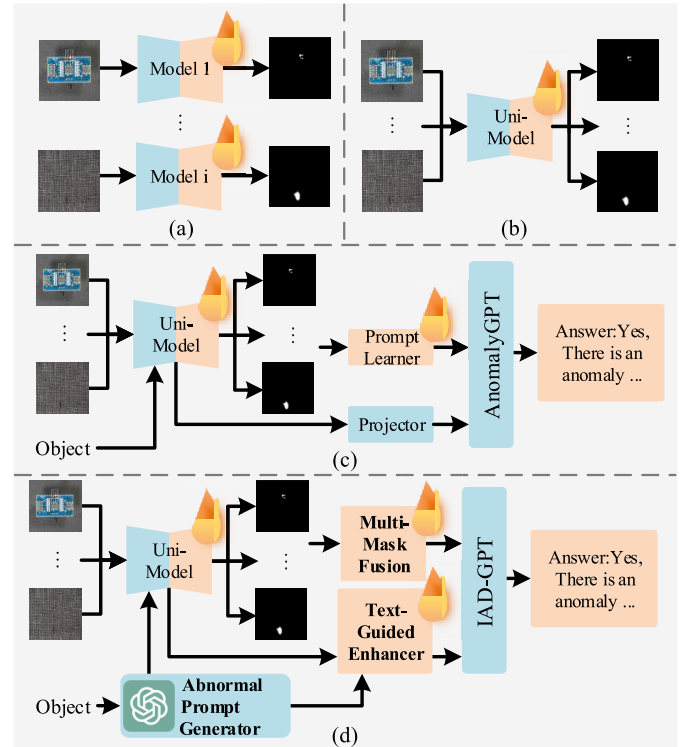


Fig. 1. Comparison between our IAD-GPT, traditional IAD methods, and AnomalyGPT. (a) Traditional methods use separate models for different classes and provide anomaly scores only. (b) Unified methods manage to accomplish anomaly detection for various classes with a unified framework. (c) AnomalyGPT, based on the settings in (b), enhances the pixel-level visual knowledge of MLLMs to perceive anomalies. (d) IAD-GPT provides GPT-generated abnormal text to improve localization capabilities and enhances image-level and pixel-level visual knowledge to achieve better anomaly recognition by MLLMs.

- 2) We employ APG to generate detailed anomaly prompts for specific objects. These prompts are utilized to activate the detection and segmentation capabilities of pretrained visual-language models via incorporating rich semantic information can significantly enhance the performance of large pretrained models in IAD tasks.
- 3) For the task of APMLLM, we design a multiscale feature enhancement approach. At image level, we develop TGE to dynamically select enhancement paths for image features. At pixel level, we introduce MMF, which leverages differences in image-text features across multiple levels to improve the LLM's ability to perceive the location of anomalies.
- 4) We achieve the state-of-the-art performance on MVTec-AD and VisA for self-supervised/few-shot anomaly detection and segmentation tasks. Compared to the baselines, IAD-GPT shows superior performance in anomaly detection and localization on images within a self-supervised learning setting, outperforming the few-shot setting.

The remainder of this article is organized as follows. Section II reviews the related works. In Section III, we describe the proposed approach in detail. Section IV presents the ablation studies and comparison experiments with the state-of-

the-art methods. Finally, Section V provides the conclusions and outlines directions for future work.

II. RELATED WORK

A. Industrial Anomaly Detection

IAD is mainly divided into reconstruction-based methods and feature embedding-based methods.

Reconstruction-based methods [1], [4], [6], [7], [19], [20], [21] rely on using only normal data when training the model, learning the feature distribution of normal data to reconstruct normal features. In the test phase, the trained model reconstructs the query data to obtain the normal feature of the query image and then compares the difference between the reconstructed image features and the original query image features to achieve the detection and location of anomalies. RealNet [7] uses a diffusion model with controllable strength to synthesize abnormal data and training the reconstruction network with abnormal data that are more similar to real-world anomalies.

Feature embedding-based [3], [5], [22], [23], [24], [25], [26], [27] methods often use networks trained on ImageNet [28] to extract features from images. The representation of abnormal areas in the image's feature space is usually far away from normal feature clusters, and anomaly detection is achieved through the obvious distance between them in the feature space. For example, PatchCore [5] constructs a memory bank storing representative patch features from normal images to detect anomalies in industrial settings without needing any examples of defects. It employs locally aware patch features aggregated from intermediate feature hierarchies of a pretrained network, ensuring spatial resolution and generality. To manage the size of the memory bank and maintain performance, PatchCore applies a coreset subsampling mechanism that selects a subset of features for efficient nearest neighbor computations. However, networks pretrained on general datasets often lack expertise in the field of IAD. Migrating general pretrained networks to specific IAD downstream tasks can make the model perform better. In SimpleNet [3], a two-layer adapter is used to transfer the features extracted from the pretrained network to the domain, synthesize abnormal data in the feature space, and then train a simple discriminator to achieve excellent anomaly detection results.

Previous studies on IAD have mainly focused on "one model for one class," and there is little research on unified anomaly detection models. UniAD [6] is a model specifically used for unified anomaly detection. UniAD uses learnable query and neighbor masking attention to prevent the model from taking shortcuts, thereby building a more robust unified anomaly detection model. DiAD [29] leverages advanced diffusion models to enhance the reconstruction and localization of anomalies across various classes. By incorporating learnable query and neighbor masking attention mechanisms in UniAD, DiAD prevents shortcut learning, thereby building a more robust model. With the powerful capabilities of pretrained visual-language models, such as CLIP, unified IAD models have more research potential. WinCLIP [8] uses the characteristics of CLIP to align images and texts and uses CLIP

for IAD tasks. Specifically, WinCLIP calculates the similarity between multiscale image features and text features representing normal/abnormal features, thereby realizing the detection of abnormal areas. AnomalyGPT [2] uses ImageBind [30] to train a simple decoder to align the feature space of images and texts to achieve IAD. By employing generalized object-agnostic text prompt templates, AnomalyCLIP [10] learns embeddings for normality and abnormality, further enhanced by global and local context optimizations to better understand anomaly semantics. AdaCLIP [9] enhances the performance of the CLIP model in zero-shot anomaly detection (ZSAD) by utilizing hybrid learnable prompts and emphasizes the importance of optimizing cues for detecting anomalies in individual images. FiLo [11] enhances the perception of anomalies in ZSAD tasks through fine-grained descriptions and high-quality localization with position enhancement.

Previous studies utilize the powerful capabilities of large pretrained models but do not fully stimulate the large pretrained models to locate anomalies at the pixel level. In this article, based on LLM and the prior knowledge of the large pretrained model, we generate possible abnormal attributes for the categories that may be encountered in the unified anomaly detection process. The specific prompts fully stimulate the capabilities of the large pretrained model, and we have achieved excellent results.

B. Multimodal LLM

With the significant progress of LLMs, such as ChatGPT and GPT-4 [13], many studies have attempted to explore other modes based on LLMs, connecting pretrained visual-language models of different modalities into end-to-end trainable models, also known as MLLMs. Due to the excellent language understanding and reasoning abilities of LLMs after large-scale data training, such as Qwen [31] and Llama [14], they have demonstrated their ability to perform translation, paraphrasing, and instruction following tasks in zero reference tasks. Models, such as MiniGPT-4 [17], Llava [16], and InstructBLIP [32], all employ fine-tuning techniques [33], [34], [35] to construct MLLMs. MiniGPT-4 [17] uses frozen Qformer and image encoder for image feature extraction based on BLIP2 [36] and trains a simple linear layer to align visual modalities into the LLM. Llava [16] is similar in architecture to MiniGPT-4, but through more diverse data and fine-tuning strategies at different stages, Llava is able to complete more complex reasoning. InstructBLIP [32] has conducted a comprehensive and systematic study on the fine-tuning of visual language instructions. The InstructBLIP model benefits from adopting a balanced sampling strategy to synchronize learning progress across datasets, enabling it to achieve excellent zero sample performance on various visual language tasks. The abovementioned multimodal big language models mainly use visual encoders pretrained on roughly aligned image text pairs, resulting in insufficient extraction and inference of visual knowledge. Therefore, more research work [15], [37], [38], [39], [40], [41] related to multimodal alignment is proposed. To address this issue, LION [39] designed a multigranularity fusion visual aggregator and used image labels as advanced semantic visual information, enabling LION to have more

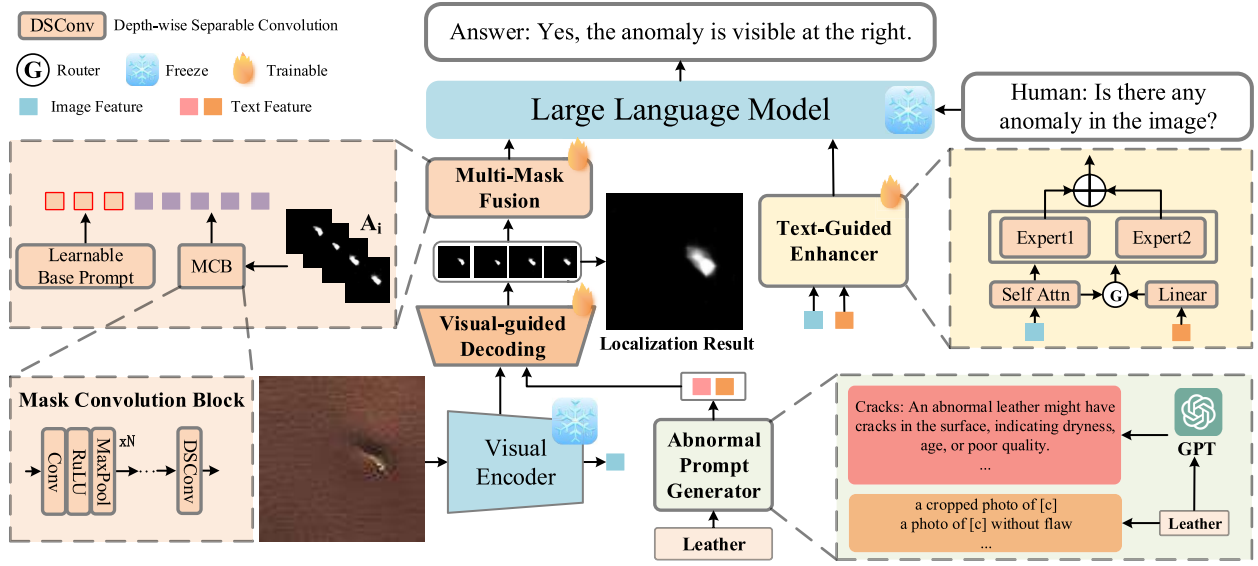


Fig. 2. Overview of IAD-GPT. APG provides category specific text prompts for decoder. TGE and MMF provide image-level visual information and pixel-level expert knowledge to LLMs, respectively.

advanced overall and fine-grained visual perception capabilities. In addition to provide visual information as an input only to LLM, methods, such as LLaMA adapter [40], [42], multimodal GPT, and Otter [15], also fuse multimodal information with intermediate features in LLMs to achieve the understanding of multimodal information by LLMs. CAT [41], [43] designed a clue aggregator to aggregate clues related to problems in dynamic audio-visual scenes, targeting rich and complex dynamic audio-visual compositions. This enriches the detailed knowledge required for learning, enabling CAT to learn clues related to problems and directly engage in action-based audio-visual reasoning. CAT outperforms other MLLMs in multimodal tasks, especially audio-visual question answering tasks.

MLLMs are trained on large-scale general datasets, which limits their capability to specifically perceive anomalies. To overcome this challenge, we introduce a method that utilizes image-level visual information and pixel-level expert knowledge. By integrating these rich sources of information, our approach significantly enhances the ability of MLLMs to perceive anomalies, thereby improving their performance in the APMLLM task.

III. METHODOLOGY

Fig. 2 illustrates the architecture of IAD-GPT. Given a query image $x \in \mathbb{R}^{H \times W \times C}$, the visual features $F_{\text{img}} \in \mathbb{R}^{1 \times C_1}$ extracted by the image encoder are passed through TGE to obtain the image embedding $E_{\text{img}} \in \mathbb{R}^{1 \times C_{\text{emb}}}$, which is then fed into the LLM.

Our method is experimentally validated in two distinct settings: a self-supervised setting, where the model learns from data with only normal samples, and a few-shot setting, which challenges the model to generalize from a very limited number of normal samples. In self-supervised setting, the patch-level features extracted by intermediate layers of image encoder are fed into the decoder together with text features that expand

anomaly prompts with APG to generate pixel-level anomaly localization results. In few-shot setting, the patch-level features from normal samples are stored in memory banks and the localization result can be obtained by calculating the distance between query patches and their most similar counterparts in the memory bank. The localization result is subsequently transformed into prompt embeddings $E_{\text{fusion}} \in \mathbb{R}^{L_1 \times C_{\text{emb}}}$ through the MMF module, serving as a part of the LLM input. The LLM detects anomalies and identifies their locations by leveraging the image input E_{img} , prompt embedding E_{fusion} , and user-provided text, thereby generating a response for the user.

A. Abnormal Prompt Generator

We design abnormal prompt generator (APG) to expand anomaly prompts to achieve more powerful segmentation capabilities. Specifically, we first prompt the LLM with the query: “describe in a paragraph what an abnormal image of $\{C_o\}$ may looks like?” with the given class C_o , and we extract potential abnormal attributes ATTR_a from the answer generated by LLM. $\text{ATTR}_a = \{k_1, k_2, \dots, k_i\}$ includes several potential abnormal keywords k_i for C_o . For each potential abnormal keyword k_i , we continue the QA session to generate a class-keyword abnormal prompt T_{k_i} . $T_{\text{app}} = \{T_{k_1}, T_{k_2}, \dots, T_{k_i}\}$ contain all potential class-keyword abnormal prompts generated by multiple rounds of dialog. For example, for leather objects, LLM is used to answer the relevant abnormal categories, including irregular texture, tears, cracks, and so on. Then, LLM is asked to generate corresponding text for each abnormal category, such as “Cracks: An abnormal leather may have cracks in the surface, indicating dryness, age, or poor quality.”

Fig. 3 shows the process of using APG to generate specific anomaly categories for leather and converting them into text prompts. We not only simply expand the anomaly categories into fixed format text but also let the LLM infer the characteristics of the object based on its own and generate

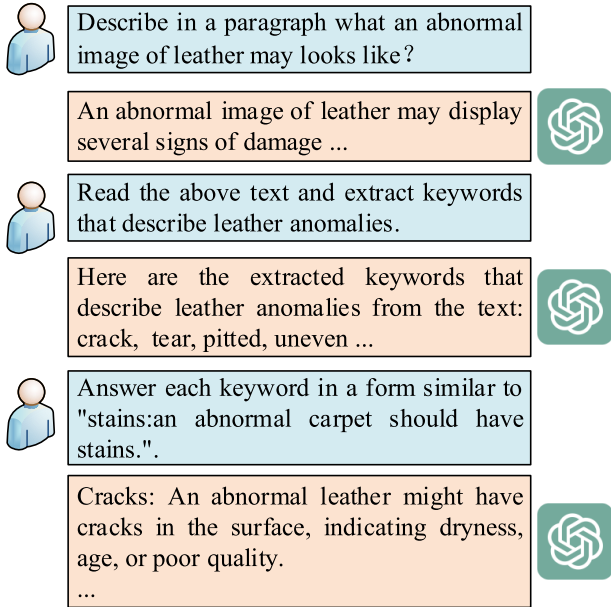


Fig. 3. Example of APG for leather. We improve the stability of LLM-generated prompts by designing an QA session and providing illustrative examples.

appropriate text prompts. WinCLIP [8] introduces a two-class design method in text prompts to help CLIP locate the abnormal region, which categorizes the text prompts into normal prompts T_n and abnormal prompts T_a , and we define text prompts similar to WinCLIP as $T_{win} = \{T_n, T_a\}$. When training the image decoder, we use T_{win} and T_{apg} as our text prompts $T_{text} = \{T_{win}, T_{apg}\}$, and we then extract the text embeddings $F_{text} \in \mathbb{R}^{L_2 \times C_2}$ using the pretrained CLIP model and align the patch-level image features $F_{patch} \in \mathbb{R}^{H \times W \times C_3}$ with the text embeddings F_{text} through a simple linear layer. The anomaly score is calculated by the similarity between the patch feature F_{patch} and text embeddings F_{text} . The localization result $M \in \mathbb{R}^{H \times W}$ can be obtained as follows:

$$M = \text{Unsample} \left(\sum_{l=1}^4 \text{Softmax} \left(\text{Linear} \left(F_{patch}^l \right) F_{text}^T \right) \right) \quad (1)$$

where l represents the number of layers. Similar to AnomalyGPT, we do not specifically select the image features of different layers for mask generation. The reason is that image features have different effects on anomaly extraction in shallow and deep layers. In previous studies, it has been found that fusing shallow and deep features helps us generate masks more accurately. For multilayer masks, we sum them up and calculate the average to obtain the final predicted mask, which is then achieved through upsampling.

For few-shot IAD, we utilize the same image encoder to extract patch-level features from normal samples and store them in memory banks $B^l \in \mathbb{R}^{N \times C_3}$. For patch-level features $F_{patch}^l \in \mathbb{R}^{H \times W \times C_3}$, the IAD localization results under the few-shot setting can be expressed as follows:

$$M = \text{Unsample} \left(\sum_{l=1}^4 \left(1 - \text{Max} \left(F_{patch}^l \cdot B^{l'} \right) \right) \right). \quad (2)$$

B. Text-Guided Enhancer

PandaGPT [44] uses a simple linear layer to align the feature space of the image encoder and LLM. However, PandaGPT has not been trained for data in the field of industrial anomalies, resulting in PandaGPT being unable to identify anomalies during IAD. Inspired by the mixture-of-experts (MoE) architecture [45], we propose the text-guided enhancer (TGE) module, in which a similar structure is designed to enhance image-level features. However, unlike the traditional MoE approach that employs a router, we dynamically control feature enhancement for each individual image through the interaction between F_{img} and F_{win}

$$W_e = \text{Softmax} \left(\text{Attn} \left(F_{img} \right) \text{Linear} \left(F_{win} \right)^T \right) \quad (3)$$

where F_{win} is the text embedding extracted by text encoder from T_{win} . We use a linear layer to align F_{win} and image-level image feature F_{img} . The enhanced image-level feature of F_{img} after self-attention is used as expert input, W_e is used as the weight of expert aggregation, and the result $E_{img} \in \mathbb{R}^{1 \times C_{emb}}$ after expert aggregation is fed to LLM as image-level feature input

$$E_{img} = \sum_{i=0}^{L_2} W_{e_i} \times \text{Expert}_i \left(F_{img} \right) \quad (4)$$

where L_2 indicates the number of categories of T_{win} , and Expert_i denotes the i th expert. Our experts are composed of a combination of an attention block and a feed-forward neural network.

C. Multimask Fusion

To utilize the masks generated by the decoder as expert knowledge and maintain semantic consistency between the LLM and the decoder output, we introduce a MMF method, which converts the localization results M_i ($i = 1, 2, 3, 4$) into a prompt embedding E_{fusion} . As shown in the left side of Fig. 2, MMF consists of multiple convolutional neural networks and trainable base prompt embeddings $E_{base} \in \mathbb{R}^{L_3 \times C_{emb}}$. Our convolutional neural network is designed to consist of multiple general convolutional layers, followed by depthwise separable convolutions. We refer to this network as mask convolution block (MCB). The MCB converts localization result M_i into prompt embeddings $E_{dec_i} \in \mathbb{R}^{L_1 \times C_3}$ and then concatenates multiple E_{dec_i} in the channel dimension to obtain an embedding $E_{fusion} \in \mathbb{R}^{L_1 \times C_{emb}}$ that fuses multilayer information. Expert knowledge $E_{expert} = \{E_{fusion}, E_{base}\} \in \mathbb{R}^{(L_1+L_3) \times C_{emb}}$ concatenates E_{base} and E_{fusion} in the length dimension and ultimately inputs them into the LLM

$$E_{dec_i} = \text{MCB} \left(M_i \right) \quad (5)$$

$$E_{fusion} = \text{Concat} \left(\left\{ E_{dec_i} \right\}_{i=1}^4 \right). \quad (6)$$

D. Data for Training

We use the NSA [46] method for training. The NSA method advances the CutPaste [47] technique by integrating the Poisson image editing [48] approach to mitigate the discontinuity

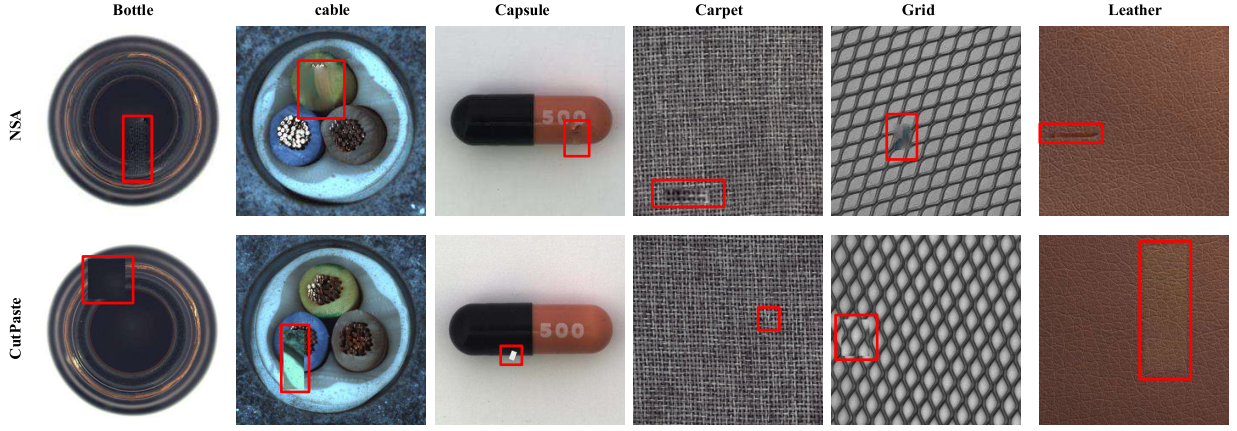


Fig. 4. Visualization comparison of anomaly image generation results between NSA and CutPaste methods. Red box indicates abnormal area.

caused by pasting image segments. In the domain of IAD, the CutPaste [47] is a prevalent method used for generating simulated anomaly images. This approach involves randomly cropping a block region from an image and pasting it onto a random location within the same or another image, thereby creating a simulated anomalous portion. While this method significantly enhances the performance of IAD models, it often results in noticeable discontinuities due to the abrupt insertion. To address these visual inconsistencies, the Poisson image editing method [48] seamlessly integrates an object from one image into another by solving Poisson partial differential equations, thereby reducing visible artifacts from direct pasting. Fig. 4 presents a visual comparison between the image results generated by the NSA method and the original CutPaste method, clearly illustrating the improvement of the NSA method in mitigating discontinuities.

In order to prevent overfitting of LLM, we use the LLM to enrich our target prompt before training LLM. For normal images, our response is designed as “No, there are no abnormalities in the image.” For abnormal images, we first generate different answer templates through LLM and define the position information of anomalies as *position*. Every time training data are generated, one of the answer templates will be selected and the position will be filled in as the answer, such as “yes, the anomaly is visible at {position}” or “yes, there is an anomaly in the image; it’s at the {position}”, and so on. For position information of anomalies *position*, we divide the image into a grid of 3×3 distinct regions to facilitate the LLM to answer the positions of anomalies, as shown in Fig. 5.

E. Loss Functions

To train our IAD-GPT, we primarily employed three loss functions: cross-entropy loss, focal loss [49], and dice loss [50]. The latter two are primarily utilized to enhance the pixel-level localization accuracy of the decoder. We only use cross-entropy loss when not training the decoder and use all three losses when training the decoder.

Cross-entropy loss is a widely used loss function for training classification models. It quantifies the difference between the predicted probability distribution and the true distribution (often represented as one-hot encoded labels). The LLM is

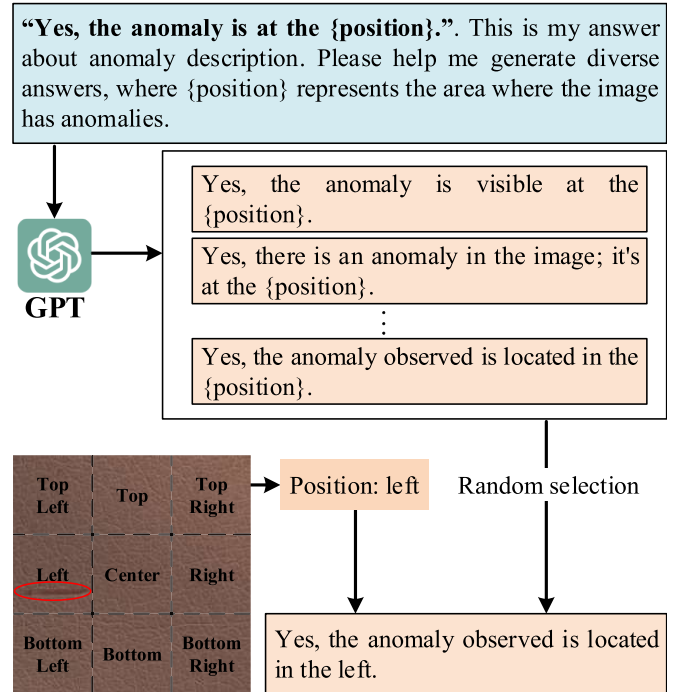


Fig. 5. Illustration of generating abnormal prompts and a 3×3 grid of images for LLM to answer abnormal locations. We first input the answer template into LLM to generate diversified answers to improve the diversity and stability of model training and then randomly select a template and fill the location information of the generated abnormal image into the answer.

trained with cross-entropy loss, which quantifies the difference between the text sequence generated by the model and the target text sequence

$$L_c = - \sum_{i=1}^n y_i \log(p_i) \quad (7)$$

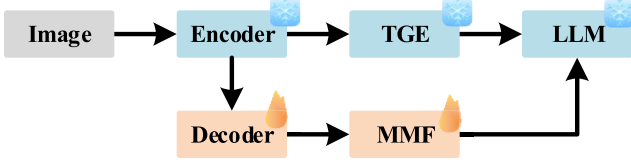
where n is the number of tokens, y_i is the true label for token i , and p_i is the predicted probability for token i .

Focal Loss is an optimized loss function specifically tailored for addressing class imbalance in classification tasks, particularly within the realm of object detection. The loss function incorporates a modulating factor (γ) that tunes down the effect of well-classified instances on the total loss, alongside an

Stage 1: Instruction-Tuning on Image-Level IAD Tasks



Stage 2: Instruction-Tuning on Pixel-Level IAD Tasks



Stage 3: Jointly-Training on Image/Pixel-Level IAD Tasks

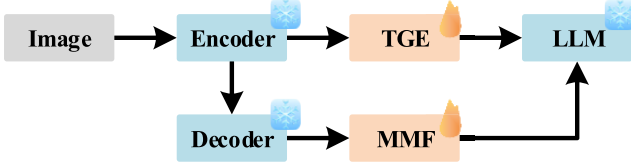


Fig. 6. Training strategy of IAD-GPT.

optional balancing factor (α) to further adjust for the disparity between classes. By doing so, focal loss enhances the model's recall on minority classes while maintaining precision

$$L_f = -\frac{1}{n} \sum_{i=1}^n \alpha_i (1 - p_i)^\gamma \log(p_i) \quad (8)$$

where $n = H \times W$ represents the total number of pixels and p_i is the probability of belonging to the true category predicted by the model. In this article, p_i is the probability of being predicted as an anomaly.

Dice loss is a performance metric turned loss function widely used in segmentation tasks to evaluate and optimize the overlap between the predicted segmentation mask and the ground truth. It measures the similarity between two samples by calculating the ratio of twice the area of intersection to the sum of the areas of the two samples. The function is particularly effective in scenarios with class imbalance due to its focus on the proportion of correctly predicted pixels relative to the total number of pixels in the target class. By minimizing dice loss during training, models are encouraged to produce segmentation outputs that have high spatial overlap with the true object boundaries, making it especially valuable for medical image analysis and other applications requiring precise boundary delineation

$$L_d = -\frac{2 \sum_{i=1}^n p_i (1 - p_i)}{\sum_{i=1}^n p_i^2 + \sum_{i=1}^n (1 - p_i)^2} \quad (9)$$

where p_i represents the probability of being predicted as an anomaly

$$L_{\text{total}} = \lambda_c \cdot L_c + \lambda_f \cdot L_f + \lambda_d \cdot L_d \quad (10)$$

where λ_f and λ_d are set to 1 in stage 2 to supervise the training of the decoder. In all other stages, these coefficients are set to 0. In contrast, the cross-entropy loss is utilized throughout all training stages, and accordingly, λ_c is set to 1 across all stages. This staged learning strategy enables the model to focus on different components of the loss function at each stage, leading to a more stable and effective training process. Further details of this training protocol are illustrated in Fig. 6.

IV. EXPERIMENT

A. Datasets

Our experiments are based on MVTec-AD [51] and VisA [52] datasets. Both benchmarks have diverse subsets of different objects, e.g., capsules and leather. In the realm of IAD, MVTec-AD stands out as a widely recognized benchmark. It contains 15 distinct categories, with a total of 3629 training images and 1725 testing images. The images within this dataset exhibit resolutions ranging from 700×700 to 1024×1024 pixels, offering a diverse array of visual data for model training. The recently introduced VisA dataset adds to the resources available for IAD research. Spanning 12 categories, it features 9621 normal images and 1200 anomalous images, with an approximate resolution of 1500×1000 pixels.

Following previous IAD methodologies, only the normal data from these two datasets are utilized during the training phase. To address the limitation of insufficient anomalous data and enable effective model training, synthetic anomalous images are generated and incorporated into the training process.

B. Evaluation Metrics

Following traditional IAD methods, we employ the area under the receiver operating characteristic (AUROC) as our evaluation metric for both detection and localization, which is expressed as image-level AUROC (I-AUROC) and pixel-level AUROC (P-AUROC). With the deployment of LLM, the existing methods allow determining the presence of anomalies without the need to manually set thresholds. We utilize image-level accuracy to evaluate the performance of our IAD-GPT.

C. Implementation Details

We use ImageBind-Huge [30] as a frozen image encoder to extract image features and Vicuna-7B [54] as LLM for reasoning, connect them through with TGE. Then, we initialize our IAD-GPT using pretrained parameters from PandaGPT [44]. We layered the training into three stages, which are stage one to train TGE, stage two to train visual-guided decoder and MMF, and stage three to train TGE and MMF jointly. At different training stages, we used the same 50 epochs on two V100 GPUs with a learning rate of 0.0005 and a batch size of 16.

Our training strategy is shown in Fig. 6. In the first stage, we do not input the mask information generated by the expert model but only train the model to better recognize the anomalous features at the image level. In the second stage, we freeze TGE on the basis of the first stage and then train visual-guided decoder and MMF, which initially aligns the pixel-level anomalous features of the mask to the feature space of the LLM. Finally, in the third stage, we freeze the visual-guided decoder and jointly train TGE and MMF to achieve a better understanding of image-level and pixel-level anomalies in the LLM.

We initialize the image as 224×224 and similar to AnomalyGPT [2], without specifying a particular level select the intermediate features of the 8th, 16th, 24th, and 32th layers

TABLE I

QUANTITATIVE RESULTS (IMAGE-LEVEL AUROC/PIXEL-LEVEL AUROC/ACCURACY) OF SELF-SUPERVISED ANOMALY DETECTION TASKS ON MVTEC-AD DATASET. WE USE BOLD AND UNDERLINE IN THE AVERAGE INDEX TO INDICATE THE BEST AND SUBOPTIMAL RESULTS, RESPECTIVELY

Method/ Category	Draem (ICCV 21)	PatchCore (CVPR 22)	SimpleNet (CVPR 23)	UniAD (NeurIPS 22)	DiAD (AAAI 24)	AnomalyGPT (AAAI 23)	IAD-GPT (Ours)
Bottle	97.5/87.6/-	100/97.4/-	97.7/91.2/-	100/96.4/-	99.7/98.4/-	99.6/94.5/97.6	99.8/98.0/100
Cable	57.8/71.3/-	95.3/93.6/-	87.6/88.1/-	95.2/97.3/-	94.8/96.8/-	89.8/86.4/83.3	93.6/91.4/88.7
Capsule	65.3/50.5/-	96.8/98.0/-	78.3/89.7/-	86.9/98.5/-	89.0/97.1/-	95.1/93.2/87.9	97.8/98.3/94.7
Hazelnut	93.7/96.9/-	99.3/97.6/-	99.2/95.7/-	99.8/98.1/-	99.5/98.3/-	99.1/91.9/94.5	100/98.7/97.3
Metal nut	72.8/62.2/-	99.1/96.3/-	85.1/90.9/-	99.2/94.8/-	99.1/97.3/-	100/94.6/100	100/98.7/100
Pill	82.2/94.4/-	86.4/90.8/-	78.3/89.7/-	93.7/95.0/-	95.7/95.7/-	94.8/84.4/88.0	94.7/97.8/88.7
Screw	92.0/95.5/-	94.2/98.9/-	45.5/93.7/-	87.5/98.3/-	90.7/97.9/-	90.2/97.3/80.6	95.4/98.9/90.0
Toothbrush	90.6/97.7/-	100/98.8/-	94.7/97.5/-	94.2/98.4/-	99.7/99.0/-	98.6/98.2/95.2	97.8/98.6/95.2
Transistor	74.8/64.5/-	98.9/92.3/-	82.0/86.0/-	99.8/97.9/-	99.8/95.1/-	96.8/75.0/92.0	88.0/85.7/79.0
Zipper	98.8/98.3/-	97.1/95.7/-	99.1/97.0/-	95.8/96.8/-	95.1/96.2/-	99.3/96.4/88.7	98.4/99.0/98.0
Object avg.	82.6/81.9/-	96.7/95.9/-	84.8/92.0 /-	95.2/ 97.2/-	96.3/ 97.2/-	96.3/91.2/90.8	96.5/96.5/93.2
Carpet	98.0/98.6/-	97.0/98.1/-	95.9/92.4/-	99.8/98.5/-	99.4/98.6/-	100/99.4/98.3	100/99.5/100
Grid	99.3/98.7/-	91.4/98.4/-	49.8/46.7/-	98.2/96.5/-	98.5/96.6/-	100 /98.2/100	100/98.8/100
Leather	98.7/97.3/-	100/99.2/-	93.9/96.9/-	100/98.8/-	99.8/98.8/-	100/99.6/100	100/99.7/100
Tile	99.8/98.0/-	96.0/90.3/-	93.7/93.1/-	99.3/91.8/-	96.8/92.4/-	99.5/97.0/98.3	99.9/99.0/98.3
Wood	99.8/96.0/-	93.8/90.8/-	95.2/84.8/-	98.6/93.2/-	99.7/93.3/-	98.8/90.9/94.9	99.8/97.9/92.4
Texture avg.	99.1/97.7/-	95.6/95.4/-	85.7/82.8/-	99.2/95.8/-	98.8/95.9/-	99.6/97.0/ 98.3	99.9/99.0/98.1
Total avg.	88.1/87.2/-	96.4/95.7/-	85.1/88.9/-	96.5/96.8/-	97.2/96.8/-	97.4/93.1/93.3	97.7/97.3/94.8

from the image encoder as input to the decoder. Linear warm-up and a one-cycle cosine learning rate decay strategy are applied. For image augmentation, the NSA [46] method is adopted, with key parameters configured as follows: Poisson image editing is implemented in normal clone mode to achieve smooth edge fusion between synthetic anomalous patches and the original image background; pixel values at the edges of patch masks are set to zero to suppress visible fusion artifacts; and the fusion center is defined as the geometric midpoint of the target pasting region in the destination image, ensuring alignment between the anomalous patch and surrounding image content. We perform alternating training using both the pretraining data of PandaGPT and our anomaly image-text data. Only TGE, visual-guided decoder, and MMF perform parameter updates at the corresponding stage, while the rest of the parameters remain frozen.

D. Self-Supervised IAD

In the setting of self-supervised training with a large number of normal samples, given that our method trains a single model on samples from all classes within a dataset, we selected AnomalyGPT [2], which is trained under the same setup, as a baseline for comparison. Additionally, we compare our model with Draem [19], PatchCore [5], SimpleNet [3], UniAD [6], and DiAD [29] using the same unified setting. The results in the MVTEC-AD dataset are presented in Table I. Our proposed method, IAD-GPT, demonstrates superior performance compared to the existing methods in most categories. We have achieved the state-of-the-art performance across multiple metrics. For Image-AUROC and Pixel-AUROC, we achieve improvements of 0.3% and 4.2%, respectively, compared to AnomalyGPT. In the task of anomaly segmentation, we demonstrated a significant improvement over AnomalyGPT,

demonstrating that APG is effective in promoting large pre-trained models to perceive anomalous features at the patch level. Among multiple multicategory anomaly detection models, our anomaly detection and localization capabilities are the best, reaching 97.7% and 97.3%. In the task of APMLLM, our accuracy rate reaches 94.8%, representing a relative improvement of 1.5% compared to AnomalyGPT.

E. Few-Shot IAD

We compare our work with prior few-shot IAD methods, selecting SPADE [53], PatchCore [5], WinCLIP [8], and AnomalyGPT as the baselines. The results are presented in Table II. Across both datasets, our method performs competitively in the IAD and APMLLM tasks and notably outperforms AnomalyGPT in the setting of 1-shot and 2-shot and achieves the state-of-the-art performance. Compared to AnomalyGPT, our method achieves better performance on Mvtec-AD and VisA for most metrics. In the one-shot and two-shot setting of the Mvtec-AD, the accuracy of IAD-GPT is 89.5% $\pm$ 1.2% and 79.1% $\pm$ 0.9%, which improves by 3.4% and 1.7% over AnomalyGPT. In other settings, IAD-GPT also achieves competitive results. This indicates that our multiscale feature enhancement approach effectively improves the LLM's ability to perceive anomalies.

In the few-shot in-context learning setting, the localization performance of the model is slightly lower than that of the self-supervised setting due to limited normal references. Our proposed use of TGE and MMF to provide multiscale anomaly perception for LLMs, which promotes the performance of LLMs in the APMLLM task. Notably, AnomalyGPT exhibits weaker anomaly localization capabilities in a self-supervised setting compared to the abilities of the model in a few-shot learning setting without training. This indicates that

TABLE II

FEW-SHOT IAD RESULTS ON MVTEC-AD AND VISA DATASETS. RESULTS ARE LISTED AS THE AVERAGE OF FIVE RUNS AND THE BEST-PERFORMING METHOD IS IN BOLD. THE RESULTS FOR SPADE, PATCHCORE, AND WINCLIP ARE REPORTED FROM [8]

Setup	Method	MVTec-AD			VisA		
		I-AUROC	P-AUROC	Accuracy	I-AUROC	P-AUROC	Accuracy
1-shot	SPADE [53]	81.0±2.0	91.2±0.4	-	79.5±4.0	95.6±0.4	-
	PatchCore [5]	83.4±3.0	92.0±1.0	-	79.9±2.9	95.4±0.6	-
	WinCLIP [8]	93.1±2.0	95.2±0.5	-	83.8±4.0	96.4±0.4	-
	AnomalyGPT [2]	94.1±1.1	95.3±0.1	86.1±1.1	87.4±0.8	96.2±0.1	77.4±1.0
	IAD-GPT (Ours)	94.1±1.1	95.3±0.1	89.5±1.2	87.4±0.8	96.2±0.1	79.1±0.9
2-shot	SPADE [53]	82.9±2.6	92.0±0.3	-	80.7±5.0	96.2±0.4	-
	PatchCore [5]	86.3±3.3	93.3±0.6	-	81.6±4.0	96.1±0.5	-
	WinCLIP [8]	94.4±1.3	96.0±0.3	-	84.6±2.4	96.8±0.3	-
	AnomalyGPT [2]	95.5±0.8	95.6±0.2	84.8±0.8	88.6±0.7	96.4±0.1	77.5±0.3
	IAD-GPT (Ours)	95.5±0.8	95.6±0.2	87.7±1.2	88.6±0.7	96.4±0.1	78.9±0.8
4-shot	SPADE [53]	84.8±2.5	92.7±0.3	-	81.7±3.4	96.6±0.3	-
	PatchCore [5]	88.8±2.6	94.3±0.5	-	85.3±2.1	96.8±0.3	-
	WinCLIP [8]	95.2±1.3	96.2±0.3	-	87.3±1.8	97.2±0.2	-
	AnomalyGPT [2]	96.3±0.3	96.2±0.1	85.0±0.3	90.6±0.7	96.7±0.1	77.7±0.4
	IAD-GPT (Ours)	96.3±0.3	96.2±0.1	84.0±0.5	90.6±0.7	96.7±0.1	78.5±0.5

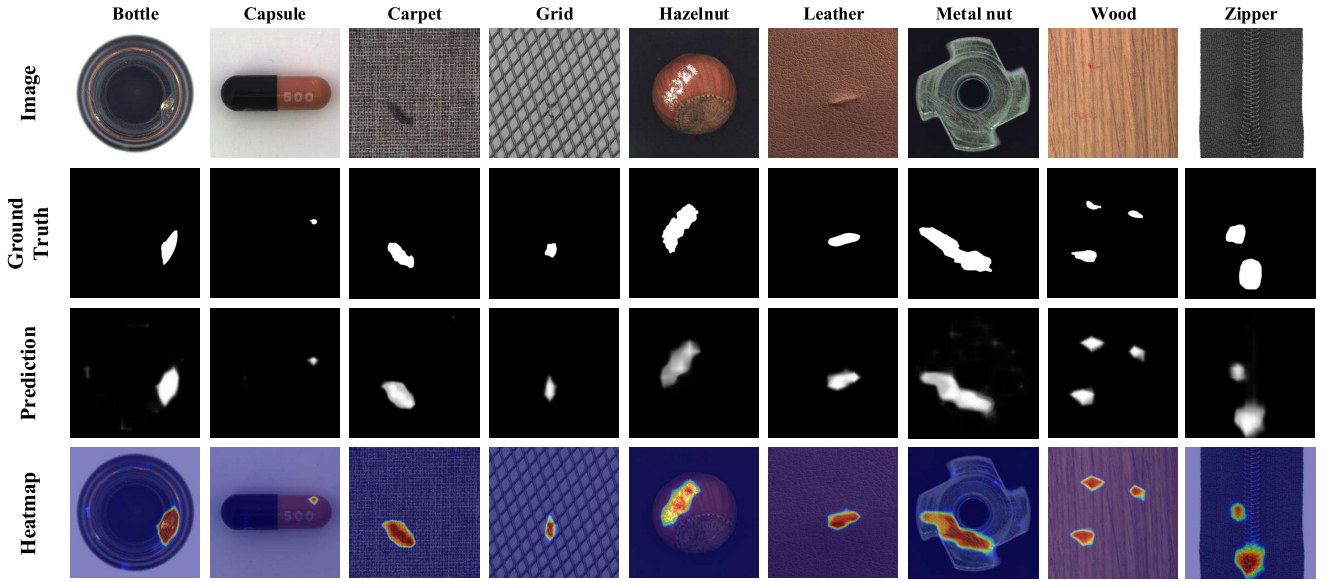


Fig. 7. Qualitative evaluation of IAD-GPT on MVTEC-AD. Input images from different categories (first row), corresponding ground truth annotations (second row), anomaly detection results predicted by IAD-GPT (third row), and prediction results using heatmaps (fourth row).

AnomalyGPT does not fully leverage the capabilities of large pretrained models. However, our proposed APG effectively compensates for this shortcoming. IAD-GPT achieves an anomaly localization performance of 97.3% P-AUROC in the self-supervised setting, surpassing the best result of 96.2% in the few-shot setting.

F. Qualitative Examples

The visualization results of IAD-GPT on the MVTEC-AD dataset can be seen in Fig. 7. It can be seen that IAD-GPT effectively identifies anomalies of different categories and has good perceptual ability in pixel-level anomaly localization. Regardless of the scale of the anomaly, whether it be large scratches or small pokes, IAD-GPT demonstrates high

accuracy in both detection and localization. Fig. 8 illustrates the performance of our IAD-GPT in self-supervised anomaly detection. Our model can not only indicate the existence of anomalies, accurately locate their locations, and provide pixel-level localization results but also answer the specific categories of anomalies that may exist, which is a capability that AnomalyGPT does not possess. Users can engage in multiturn conversations related to the image content, including but not limited to asking IAD-GPT whether the image contains anomalies or requesting specific descriptions about the image.

G. Ablation Study

To evaluate the effectiveness of each proposed module, extensive ablation experiments were conducted on the MVTEC-

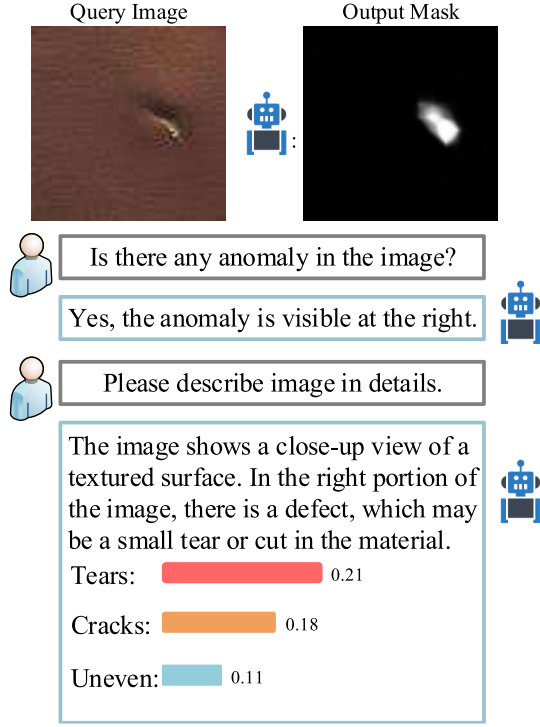


Fig. 8. Qualitative example of IAD-GPT on MVtec-AD. Anomaly categories are computing from the similarity between F_{img} and F_{apg} .

TABLE III

ABLATION OF TGE IN DIFFERENT FRAMEWORKS. ACC. DENOTES ACCURACY (%)

TGE	IAD-GPT	AnomalyGPT	I-AUROC	P-AUROC	Acc.
			-	-	72.2
✓			-	-	82.3
		✓	97.3	93.1	93.3
✓		✓	97.3	93.1	93.6
✓	✓		97.7	97.3	94.0

TABLE IV

ABLATION STUDY ON THE INTEGRATION OF EXPERT KNOWLEDGE INTO LLM

APG	MMF	Prompt Learner	I-AUROC	P-AUROC	Acc.
		✓	97.3	93.1	93.3
✓		✓	97.5	95.6	93.6
✓	✓		97.7	97.3	94.0

TABLE V

COMPARISON OF PROMPT LEARNER FROM ANOMALYGPT AND MMF FROM IAD-GPT ON THROUGHPUT (IMGS/S), PARAMETERS (M), FLOPS (G), AND ACCURACY (%)

Module	Throughput↑	Parameters↓	FLOPs↓	Acc.↑
Prompt Learner	97.8	107.4M	15.6G	93.3
MMF	114.2	10.2M	49.3G	94.8

AD dataset. Our study primarily focuses on three key aspects: the TGE, the integration of expert knowledge, and the multistage training strategy. The main results are summarized in Tables III–VII. All analyses are based on self-supervised training and testing protocols applied to the MVtec-AD dataset.

TABLE VI
ABLATION OF TRAIN STRATEGY

Multi-stage	IAD-GPT	Anomaly-GPT	I-AUROC	P-AUROC	Acc.
		✓	97.3	93.1	93.3
	✓		97.7	97.3	94.0
✓	✓		97.7	97.3	94.8

TABLE VII
COMPARISON OF IMAGE AUGMENTATION METHODS

Method	I-AUROC	P-AUROC
NSA [46]	97.7	97.3
CutPaste [47]	92.1	89.2
NSA [46] + CutPaste [47]	94.1	91.1

1) *Impact of TGE*: To demonstrate the effectiveness of the TGE in enhancing visual information, we train the model for anomaly perception using only the TGE. As shown in Table III, compared to PandaGPT, our approach achieves a performance improvement of 10.1%. To further validate the applicability of the TGE across different frameworks, we also conducted ablation studies on AnomalyGPT. The experimental results confirm that the TGE consistently improves the model's ability to perceive anomalies at the image level, thereby enabling better performance on APMLLM task in both IAD-GPT and AnomalyGPT.

2) *Impact of Expert Knowledge*: To demonstrate the impact of the expert knowledge incorporated via APG and MMF, we compare the performance of AnomalyGPT with our method after integrating expert knowledge. As shown in Table IV, APG consistently improves both anomaly detection and localization across different frameworks, indicating that APG is effective in promoting large pretrained models to perceive anomalous features at the patch level.

To enable the LLM to better comprehend and utilize the expert knowledge, we propose the MMF module. Unlike the prompt learner used in AnomalyGPT, MMF fully exploits multilevel expert knowledge during the prompting process. In Table V, we compare the efficiency of MMF and the prompt learner. MMF achieves superior performance in terms of throughput and parameter count, reaching 114.2 imgs/s and 10.2M parameters, compared to 97.8 imgs/s and 107.4M parameters for the prompt learner. However, due to the additional overhead of processing multilayer expert knowledge, MMF incurs a higher computational cost, as reflected by its significantly larger FLOPs. In terms of accuracy, IAD-GPT achieves 94.8%, outperforming AnomalyGPT's 93.3%, demonstrating the effectiveness of our design.

3) *Impact of Training Strategy*: To evaluate the effectiveness of the multistage training strategy, we present its impact on IAD-GPT in Table VI. Without multistage training, our method still outperforms AnomalyGPT across all evaluation metrics. The incorporation of multistage training further enhances IAD-GPT's ability to perceive anomalies in APMLLM task, leading to improved performance in both detection and localization.

4) *Impact of Data Augmentation*: We have supplemented a more detailed comparative ablation experiment focusing on data augmentation methods, specifically evaluating three scenarios: training with only the NSA-based augmentation, training with only the CutPaste augmentation, and training using a combination of NSA and CutPaste. All experiments strictly followed the experimental setup in Section IV-C. For the combination scheme, we randomly selected either NSA or CutPaste to synthesize anomalous images in each training iteration before feeding them into the model.

The experimental results in Table VII show that the NSA-based augmentation achieves the best performance in both anomaly detection and localization tasks. We believe this is attributed to its ability to generate more realistic anomalous regions with smoother edge transitions, which helps the model learn more discriminative normal–abnormal feature differences.

V. CONCLUSION

In this study, we introduce IAD-GPT, an innovative framework for IAD. IAD-GPT leverages the advanced capabilities of MLLMs and integrates multiscale visual information through TGE and MMF. TGE effectively enhances the alignment between image-level visual information and LLMs and improves LLM’s perception of anomalies by dynamically selecting enhancement paths for image features. Meanwhile, the MMF module integrates multilevel localization results as visual expert knowledge for LLM to enhance its pixel-level anomaly perception. Our experiments on benchmark datasets, such as MVTec-AD and VisA, highlight the superior performance of IAD-GPT. IAD-GPT achieves better performance in APMLLM task by leveraging multiscale visual information. Furthermore, it fully enhances the capabilities of large pre-trained models based on APG to detect and localize image anomalies. We have improved our performance in anomaly detection and localization compared to the baseline, and due to the excellent performance of APG, we have achieved better anomaly localization performance in the self-supervised setting than in the few-shot in-context learning setting.

IAD-GPT provides a more comprehensive and robust LLM-based solution for industrial applications. Beyond its technical contributions, this work also underscores the broader potential of leveraging MLLMs in industrial domains, opening new avenues for interactive and explainable artificial intelligence solutions. Future work will explore the extension of IAD-GPT to other fields, such as medical anomaly detection and camouflage object detection. In addition, efforts will be made to improve its adaptability to more complex industrial scenarios.

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